Privacy by Choosing Less

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach privacy through concrete information choices.

Model workshop and tool-counting plans with fictional data, then reserve the library case for independent evidence.

Keep one habit: Look for purpose-based necessity, realistic access and honest deletion limits.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Justify which information a stated task needs and remove unnecessary fields

Teach **privacy** as **purpose**ful information design, not a demand that pupils disclose private experiences. Use fictional paper examples throughout; no account creation, real survey or upload is necessary.

These examples are classroom design exercises, not legal advice or a substitute for school policy. Coded records with a re-identification key remain linkable.

Aggregation reduces individual detail but does not guarantee anonymity in every context.

By the end, I can collect only information necessary for a stated task and explain coded-record limits.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; **personal data** and paid accounts are unnecessary.
- Fictional topic slips or counters marked drawing, music and coding
- A sample form containing only the fictional field labels in W1
- Paper for the five-question data plan; no real pupil or borrower records

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a correct-out-of-assessed fraction and explain its denominator

Reference example, not the only acceptable wording: A has 9 correct out of 12 assessed: 9/12 = 3/4 = 75%. The denominator includes all 12 judged outcomes, including 3 incorrect ones.

Using 9/9 would exclude the errors and falsely report perfect assessed accuracy.

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

Look for: Check records against a stated format, unit and meaning before using them

Reference example, not the only acceptable wording: Height 60 fails the whole-number range 1 through 10. Travel blank is missing; its value cannot be inferred.

Travel 0 is valid in the whole-number range 0 through 10 and can represent observed no travel. Zero is a measurement, not a replacement for a blank.

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a confidence claim from **evidence** that a **prediction** was correct

Reference example, not the only acceptable wording: The learner replaced the documented score meaning with a probability interpretation. Report 80 of 100 feature-match points.

To claim a chance of correctness, a defined probability method and relevant outcome evaluation would be needed.

No. It needs the assessed outcomes and counts, not a student's identity or home address. The required information depends on the **purpose**.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Fair for Whom?

Good Data, Bad Data

Confidence Is Not Correctness

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A book recommender requests genre, maximum pages, home address and a selfie. Keep only fields needed for genre and length matching. Explain one consequence of storing extra fields.

Concept to explain

A useful system should collect only information it truly needs.

Term	Meaning
privacy	control and protection of information about people

Term	Meaning
data minimization	collecting only what the purpose requires
purpose	specific reason information is used
retention	how long information is kept

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach purpose before fields, access and retention

> Use fictional cases to distinguish minimisation, coded records and justified changes in purpose.

Model the workshop purpose and rejected fields

Privacy includes having appropriate boundaries around information about people. A useful first question is: what exact job are we trying to do? A second is: what is the least information that can do that job?

Collecting extra details just because a form has room for them creates **more information** to protect.

Our fictional class wants to choose one of three workshop topics for Friday: drawing, music or coding. It needs a total for each topic, not a list of who chose what.

Each fictional pupil gets one paper slip and marks one topic. No real class survey is needed for this lesson. The teacher counts the choices and uses the largest total; a tie goes to a class discussion.

A proposed form asks for name, full date of birth, home address, photograph and topic choice. For the stated counting task only the topic choice is needed. Remove the other fields before collection.

Do not fill them in and promise to ignore them later. If the **purpose** changes to individual attendance, design and justify that separate task with the teacher rather than silently reusing the survey.

Check totals and specify retention

Suppose the fictional counts are drawing 8, music 7 and coding 5. These totals support the choice of drawing. There is no need to publish each slip or a photograph of a child holding it.

The teacher checks the total of 20 against the number of slips and then follows the agreed school **process** for disposing of the slips after verification.

A **retention** plan needs a clear trigger and a responsible person: for example, the teacher disposes of the fictional vote slips after the totals are checked and the activity closes.

Real classroom records must follow the school's approved **process**. Deleting one visible copy does not prove that every backup, screenshot or previously shared copy is gone.

Inspect coded records, photos and small groups

Proposed change	What it changes	What still needs thought
Replace a name with P07	Hides the name on this sheet	A separate key can reconnect the code
Remove names but keep photos	Removes one identifying field	Faces may still identify people
Share topic totals only	Removes individual rows	Small or unusual groups may still reveal a choice

A code is not a promise of anonymity. If a teacher keeps a key connecting P07 to a pupil, the coded record can still be linked to that pupil.

Even without a key, a combination such as an exact birthday, a team and a photograph may identify someone. Ask whether a reader could connect the information to a person, not just whether the name column is blank.

Totals also need context. A report saying that the only member of a tiny club chose music reveals that member's choice.

Our whole-class topic totals reveal less detail than individual rows, but we should not describe every possible total as automatically anonymous. Share only what the audience needs for the **purpose**.

Return to the workshop vote's information journey

AI ACADEMY / AI-1/24 · 03 · INSPECT THE LESSON MODEL

A small information plan for a fictional vote

Stage	Needed action	Privacy check
Collect	One topic per slip	No name or address
Count	Teacher totals choices	Keep slips restricted
Share	Three topic totals	No individual choices
Delete	After totals checked	Follow agreed plan

Purpose: choose a workshop topic. Collect the minimum, limit access, share the required result and agree how long the source slips are kept.

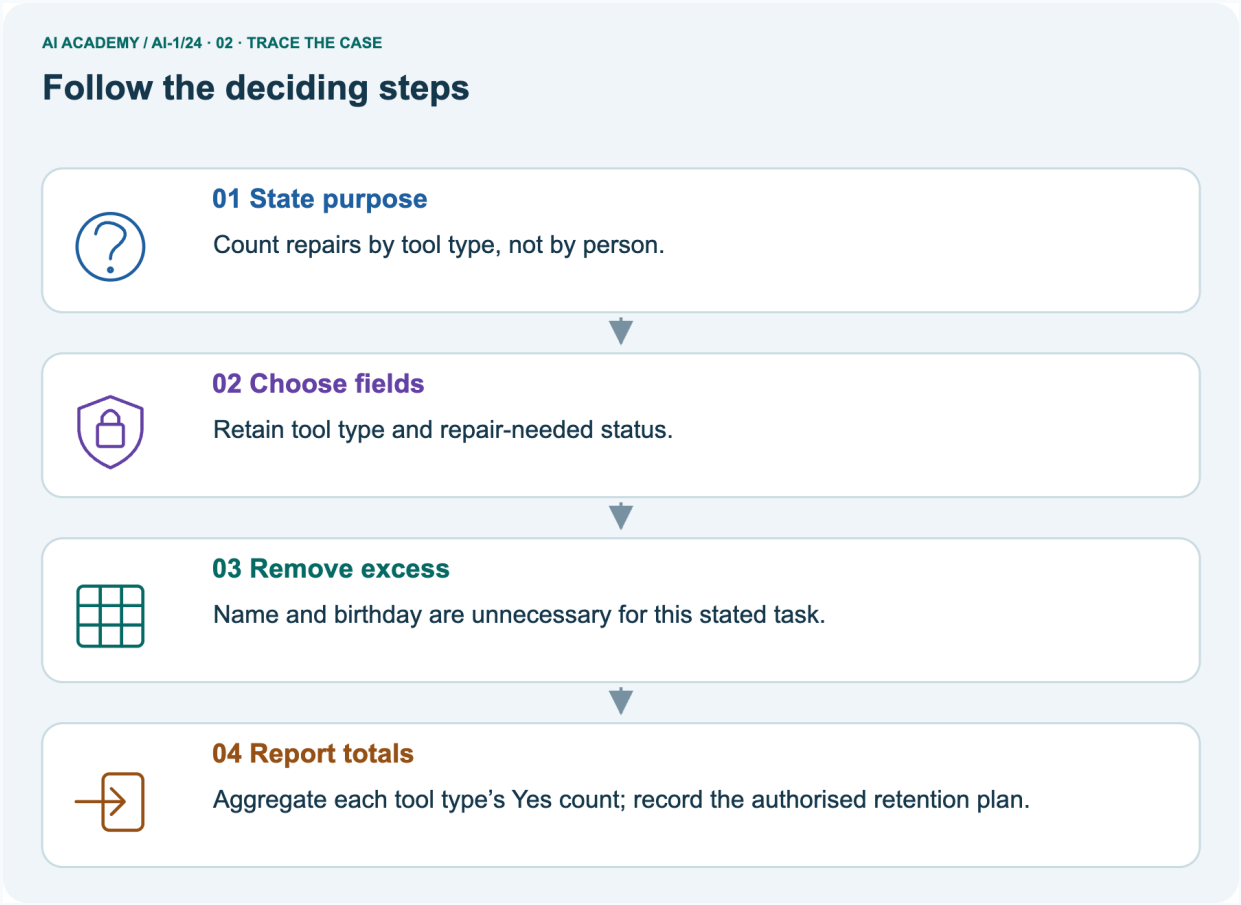
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the separate tool-repair case

A fictional teacher wants to count broken craft tools by type, without contacting borrowers.

Needed **output**: totals for scissors, ruler and hole-punch repairs.

Draft form requests pupil name, birthday, tool type and repair-needed Yes/No. No borrower follow-up is required.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State purpose	Count repairs by tool type, not by person.
2. Choose fields	Retain tool type and repair-needed status.
3. Remove excess	Name and birthday are unnecessary for this stated task.
4. Report totals	Aggregate each tool type’s Yes count; record the authorised retention plan.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals.

Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

Limit anonymity claims

Removing these fields is not a universal guarantee of anonymity or permission to collect other identifying combinations.

Reassess an authorised changed task

Change: The teacher adds an authorised task to return one repaired tool to its borrower.

Prediction: A minimal contact **mechanism** may now be needed for that tool.

Reason: The **purpose** changed; reassess necessity rather than restoring every old field.

Connect the pictures to decisions

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT** candidate information fields

> **2. Process — the deciding step**
>
> **PROCESS** keep only fields needed for the stated goal

> **3. Output — the result**
>
> **OUTPUT** useful result with reduced **privacy** risk

CORE IDEA A useful system should collect only information it truly needs.

UNDER THE HOOD **Data minimization** limits collection, access, **retention**, and reuse to what is necessary for a defined **purpose**.

Less data reduces breach exposure and misuse, but essential high-stakes information still needs secure, consent-aware handling.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A lunch helper needs food preferences and allergy information for its task, not a face photo, home address, or family income.

STEP 1 — OBSERVE candidate information fields

STEP 2 — TRACE keep only fields needed for the stated goal

STEP 3 — CONCLUDE useful result with reduced **privacy** risk

MODEL REASONING Keep only information required for safety and stated preferences, allow a person to provide or update it, restrict access, and delete it when the **purpose** ends.

Explain why each removed field adds risk without benefit.

DO NOT OVERCLAIM Useful does not mean permitted forever or for every new **purpose**.

What new **evidence** could change this answer?

Concrete teaching case

A genre-based book helper asks for genre, home address and a photograph.

Which fields are necessary?

Exact explanation to model

Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

Explain the added worked example

Key terms

Term	Meaning
Personal data	Information about an identifiable person, including combinations that can identify someone

Term	Meaning
Purpose	The specific job information is collected or used to do
Data minimisation	Using only the information needed for a stated purpose
Access control	A rule and mechanism deciding who can see or use information
Retention	How long information is kept before an agreed deletion or other approved action

Model the reasoning aloud

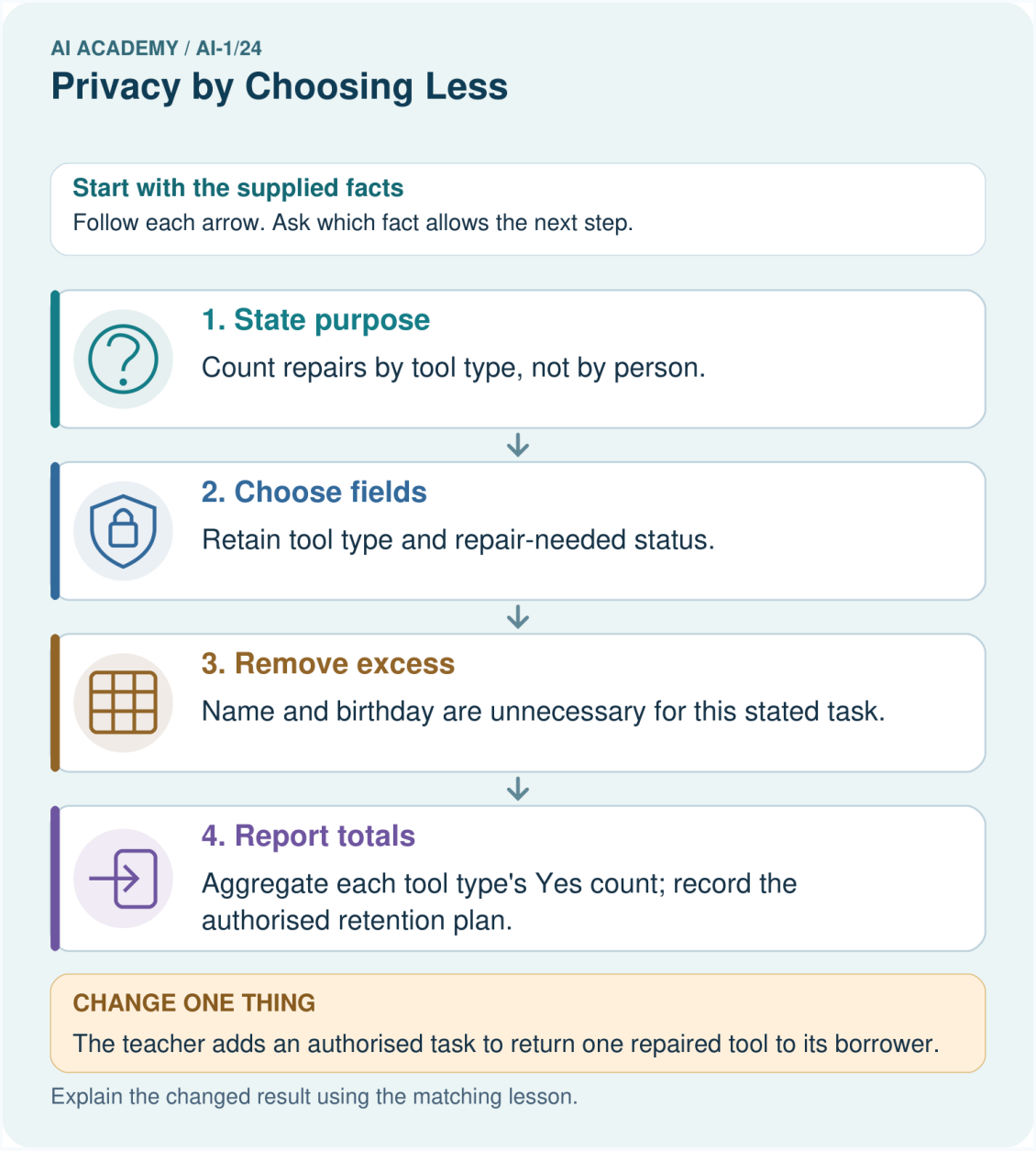
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Privacy by Choosing Less

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: A teacher replaces names with codes and keeps a separate code-to-name key. A pupil calls the records anonymous. Explain the problem.

Then explain why removing names while leaving photographs would not settle the question either.

Expected reasoning and fading help

Teacher check: The key can reconnect each code to a person, so the coded records are not anonymous just because names are absent. Restricting the key helps limit access but does not erase the link.

Photographs can identify people even if the name field is removed.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Privacy** and **data minimisation**. Proposed explanation: “More data always improves a task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Unnecessary identity fields do not improve a category count and create **more information** to manage.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

For the original task, does adding an anonymous-looking pupil code improve the repair count?

No. The totals need tool type and repair status, not pupil identity. A linkable code can also retain personal-data risk.

Changed-case reasoning: A minimal contact **mechanism** may now be needed for that tool. The **purpose** changed; reassess necessity rather than restoring every old field.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ **Earlier wording, alternatives and picture transcripts**

Demonstrate and guide practice

Board sequence

Write the given facts: A genre-based book helper asks for genre, home address and a photograph.

Write the question to explain: Which fields are necessary?

Reveal after **prediction**s: Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

Use the earlier changed question as optional later practice. Reserve Design a different minimal-data task for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

A chart needs the three topic totals, not the named spreadsheet. Use the fictional counts provided here.

Before any real tool use, check teacher approval for both tool and data; a smaller **input** does not itself authorize an upload. Draw bars on paper or use an approved local chart tool.

Exit: collect topic choice; remove any two identity/contact fields. Share totals with the class. The teacher disposes of slips after verification and task closure under the agreed school **process**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve which fields a performance count actually needs. Establish fictional examples only.
7 to 20 minutes	Define the vote purpose. Complete W1 and remove unnecessary fields before collection.
20 to 33 minutes	Trace collection, access, sharing and disposal on the diagram. Check the totals.
33 to 45 minutes	Complete W2. Contrast coded records with a claim of anonymity.

Stage	Teacher action
Session 2 start 0 to 8 minutes	Retrieve why a blank name column does not prove anonymity.
8 to 23 minutes	Complete W3 and the guided chart task. Discuss small-group inference and approved tool rules.
23 to 38 minutes	Collect W4 independently. Check that purpose, fields, access, output and disposal align.
38 to 45 minutes	Complete W5 and the exit. Rehearse telling a trusted adult without repeating exposed information.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the fictional lunch-counting redesign, retain only menu choice for the stated purpose. Rice 9 + pasta 7 + soup 4 = 20. Name, birthday, address, photograph and favourite colour do not improve those totals. Model restricted handling by the teacher, sharing only the required totals, and disposal by an identified responsible person after agreed verification and activity closure under the approved process. The totals-only plan does not manage individual dietary

safety: an authorised changed purpose requires a separately reviewed process. Do not collect real health or pupil data for this activity. Reserve Design a different minimal-data task for the independent first attempt.

Run and record

1. Complete the task: Redesign a lunch helper that asks for too much **personal data**.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

More data always improves a task: Unnecessary identity fields do not improve a category count and create **more information** to manage.

Codes or missing names mean anonymous: A key or other identifying fields may reconnect records to a person.

Every total is safe to publish: A small or distinctive group may allow a reader to infer a person’s choice.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Design a different minimal-data task for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR Collecting extra personal details always improves a tool.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Deleting one copy proves all copies are gone: Other copies, screenshots or backups may remain; follow the adult-led school **process**.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A fictional library wants only the total number of books needing repair in each category: cover, page or binding. It does not need to contact borrowers.

A draft repair sheet asks for borrower name, home address, book repair category and borrower photograph.

Design a plan: state the **purpose**, required and removed fields, who handles source sheets, what totals are shared with the librarian, and who disposes of sheets at what agreed point. Use no real borrower details.

Purpose: count repair needs by category. Keep repair category; remove borrower name, address and photograph because no borrower contact is required.

An assigned teacher or librarian handles the source sheets, checks counts and shares category totals with the librarian.

Name that adult as responsible for disposal after totals are verified and the counting task closes, following the agreed school **process**. Other justified restricted roles and clear disposal triggers are acceptable.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trim a form to its purpose

The fictional workshop vote asks for name, full date of birth, address, photograph and topic choice. Its only **purpose** is to count votes for drawing, music and coding. State which field to keep, which to remove and why.

Explain why collecting the extra fields first is different from never collecting them.

Keep topic choice; remove name, full date of birth, address and photograph. The totals do not require identity or contact details.

Collecting extra fields creates copies and exposure to manage even if those fields are later ignored or deleted.

W2 Challenge a claim of anonymity

A teacher replaces names with codes and keeps a separate code-to-name key. A pupil calls the records anonymous. Explain the problem.

Then explain why removing names while leaving photographs would not settle the question either.

The key can reconnect each code to a person, so the coded records are not anonymous just because names are absent. Restricting the key helps limit access but does not erase the link.

Photographs can identify people even if the name field is removed.

W3 Share the result the audience needs

The fictional class has drawing 8, music 7 and coding 5. Write the result needed to choose a workshop. A pupil suggests posting every vote slip online as proof. Explain a lower-exposure way to check and share the result.

Why can a total about a one-person club still reveal a choice?

Drawing leads with 8; totals are 8, 7 and 5, adding to 20. The teacher checks the slips and totals, then shares the three totals with the class rather than public individual slips.

A one-person club total can be linked to its only member, revealing that person's choice.

W5 Respond to an unnecessary upload

Someone plans to upload the library repair sheet with borrower details to an unapproved AI tool to add three totals. Explain a smaller-**input** alternative and a no-upload method.

If the sheet was already shared, state a suitable first response and one claim you must not make about deleting it.

Use only fictional category counts in this lesson; for real work check teacher-approved tools and data rules before any upload. Add totals on paper or with an approved local calculator.

If shared, stop further sharing and tell the teacher or trusted adult promptly so the school **process** can be followed. Do not claim that deleting the visible message removes all copies or backups.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Genre is relevant; address and photo are unnecessary for the stated task. Collecting less reduces exposure.

Additional transfer practice question

A book helper needs reading topic and approximate difficulty. Someone asks it to collect home address too. Which information can be left out, and why?

Reasoned practice answer

Leave out home address because it is unnecessary for the stated recommendation task. Collect only information that serves the learning **purpose**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Collecting extra personal details always improves a tool.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Keep genre and maximum pages. Address and selfie do not help the stated task and create unnecessary exposure if shared or lost. Use fictional records for practice and specify who may see needed data and when it is deleted.

Explanations for the additional workbook examples

W1 Trim a form to its purpose

W2 Challenge a claim of anonymity

W3 Share the result the audience needs

W4 Design a different minimal-data task

A fictional library wants only the total number of books needing repair in each category: cover, page or binding. It does not need to contact borrowers.

A draft repair sheet asks for borrower name, home address, book repair category and borrower photograph.

Design a plan: state the **purpose**, required and removed fields, who handles source sheets, what totals are shared with the librarian, and who disposes of sheets at what agreed point. Use no real borrower details.

Purpose: count repair needs by category. Keep repair category; remove borrower name, address and photograph because no borrower contact is required.

An assigned teacher or librarian handles the source sheets, checks counts and shares category totals with the librarian.

Name that adult as responsible for disposal after totals are verified and the counting task closes, following the agreed school **process**. Other justified restricted roles and clear disposal triggers are acceptable.

W5 Respond to an unnecessary upload

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Cross out names, faces, and ownership details unless an explicit need survives review; set a deletion rule for sorter records.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Write the sorter's **purpose**, necessary fields, access rules and deletion point before collecting any real classroom information.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Offer a five-box planning frame: **purpose**, fields, access, shared result, disposal. Accept spoken explanations of why each field is or is not needed. Avoid asking children to describe past **privacy** incidents.

Extension: construct a fictional small-group total that reveals one choice and suggest a less detailed **output**.

W4 requires a category-count **purpose**, repair category only from the listed fields, removal of all three borrower fields, a restricted responsible adult, category totals for the librarian and an actionable disposal trigger under the agreed **process**.

After support change the task to contacting borrowers about repairs.

Ask the pupil to explain that the old minimal field set no longer performs the new task and that an adult must design an approved contact workflow; do not request real contact details.

Plan the delayed check

Suggested return points in this level: ai-1/25, ai-1/27, ai-1/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 25: Generate, Retrieve, or Calculate?. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 25 distinguishes generating a plausible response, retrieving an existing record and calculating a result from explicit rules.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Justify which information a stated task needs and remove unnecessary fields

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: A teacher replaces names with codes and keeps a separate code-to-name key. A pupil calls the records anonymous. Explain the problem.

Then explain why removing names while leaving photographs would not settle the question either.

Reasoned practice answer: The key can reconnect each code to a person, so the coded records are not anonymous just because names are absent. Restricting the key helps limit access but does not erase the link.

Photographs can identify people even if the name field is removed.

Check every learner before discussion: Use the worked case for **Privacy** and **data minimisation**. Proposed explanation: “More data always improves a task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Unnecessary identity fields do not improve a category count and create **more information** to manage.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The teacher adds an authorised task to return one repaired tool to its borrower. Require a **prediction** and a reason before comparing. Expected result: A minimal contact **mechanism** may now be needed for that tool.

Reason: The **purpose** changed; reassess necessity rather than restoring every old field.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

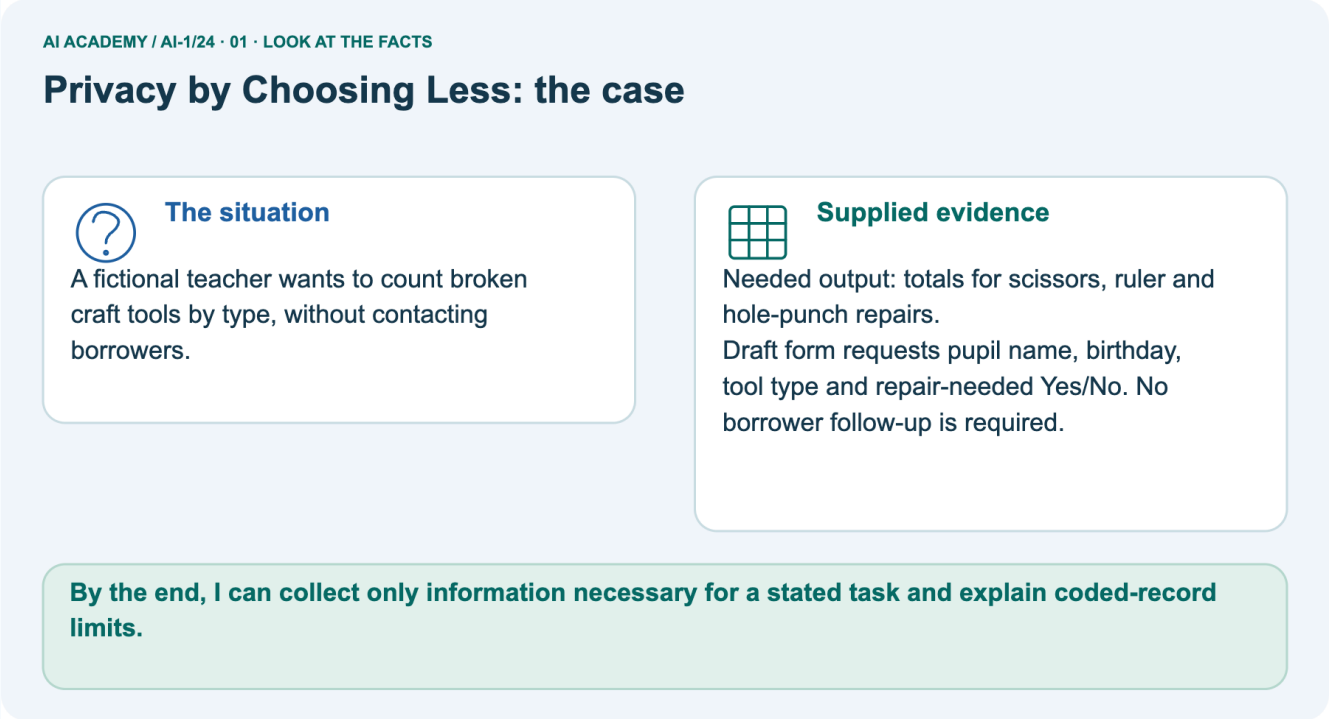
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Privacy by Choosing Less: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

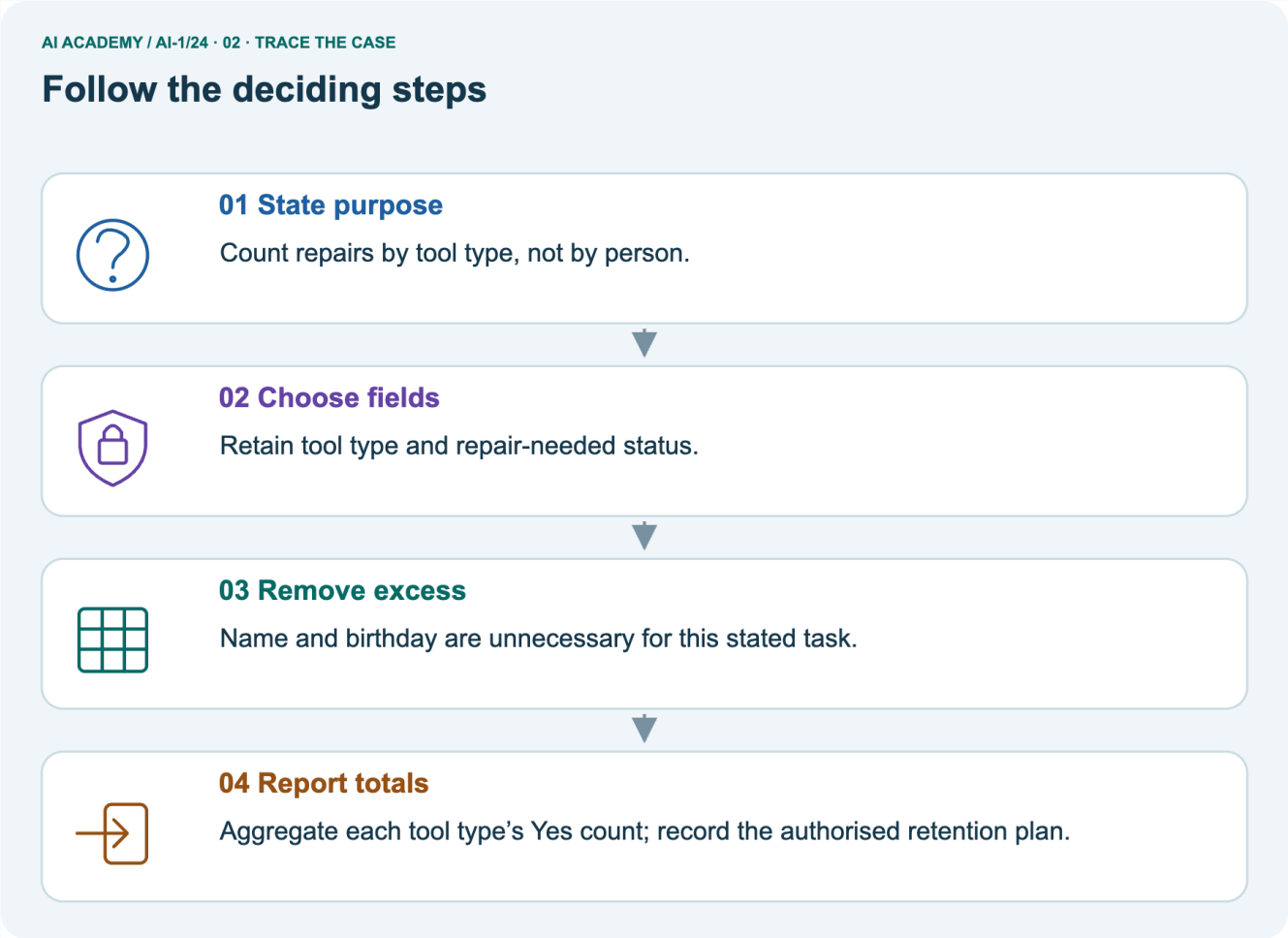
A fictional teacher wants to count broken craft tools by type, without contacting borrowers. Supplied facts: Needed **output**: totals for scissors, ruler and hole-punch repairs.

Draft form requests pupil name, birthday, tool type and repair-needed Yes/No. No borrower follow-up is required.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



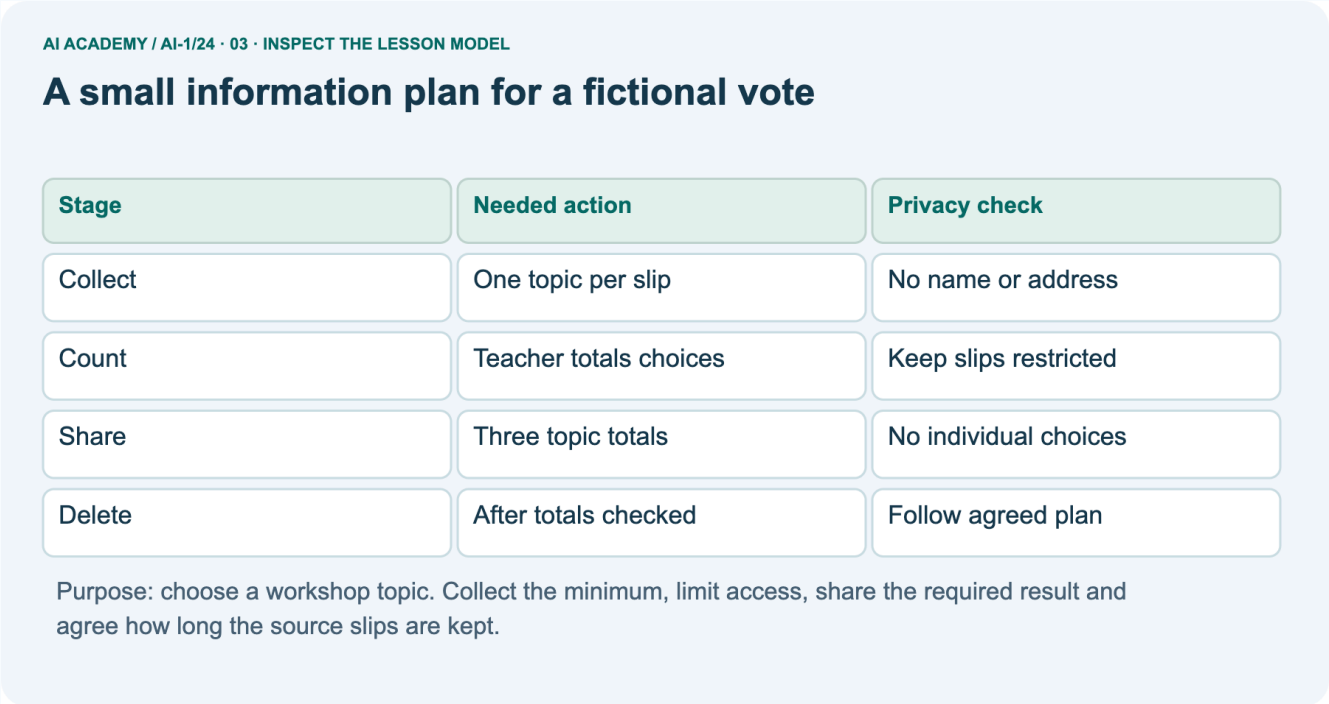
Follow the labelled picture with the matching explanation. Open the vector diagram

State **purpose**: Count repairs by tool type, not by person. → Choose fields: Retain tool type and repair-needed status. → Remove excess: Name and birthday are unnecessary for this stated task. → Report totals: Aggregate each tool type's Yes count; record the authorised **retention** plan.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A small information plan for a fictional vote



Follow the labelled picture with the matching explanation. Open the vector diagram

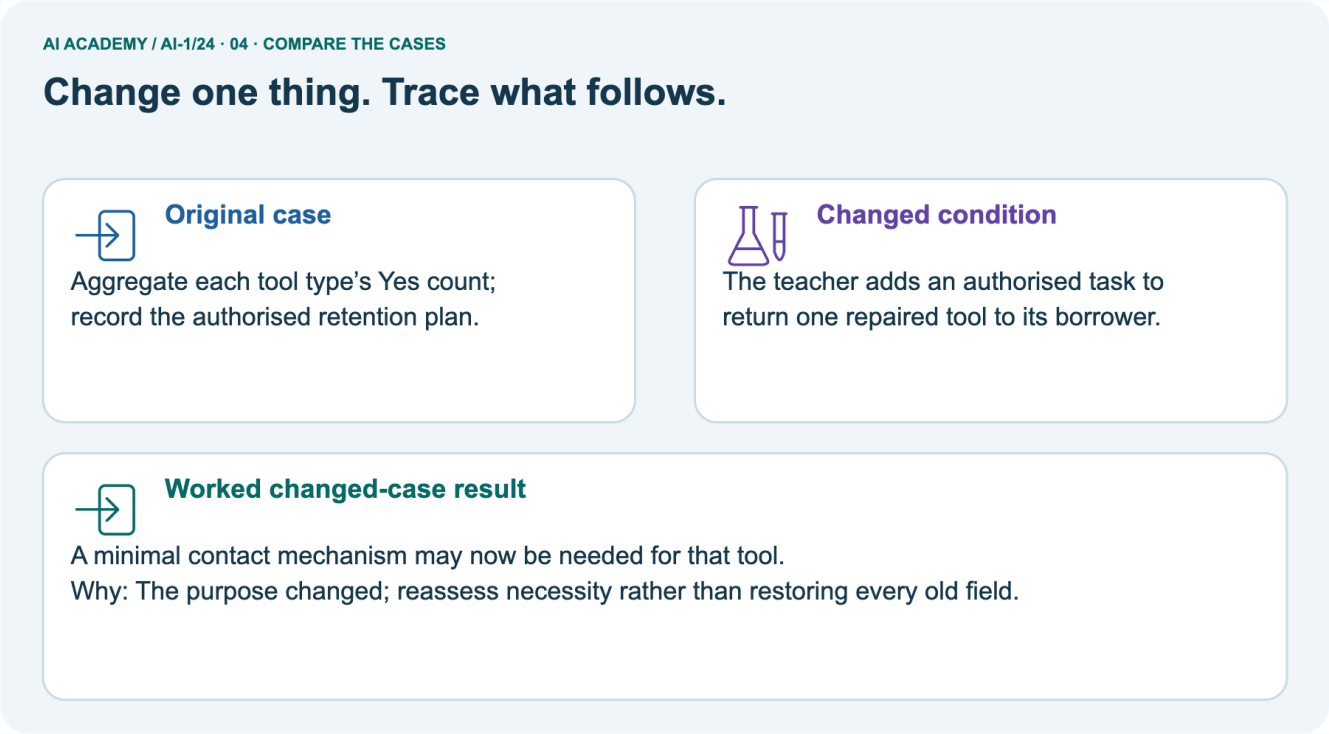
A small information plan for a fictional vote. Collect | One topic per slip | No name or address. Count | Teacher totals choices | Keep slips restricted. Share | Three topic totals | No individual choices.

Delete | After totals checked | Follow agreed plan

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Aggregate each tool type's Yes count; record the authorised **retention** plan. Changed condition: The teacher adds an authorised task to return one repaired tool to its borrower.

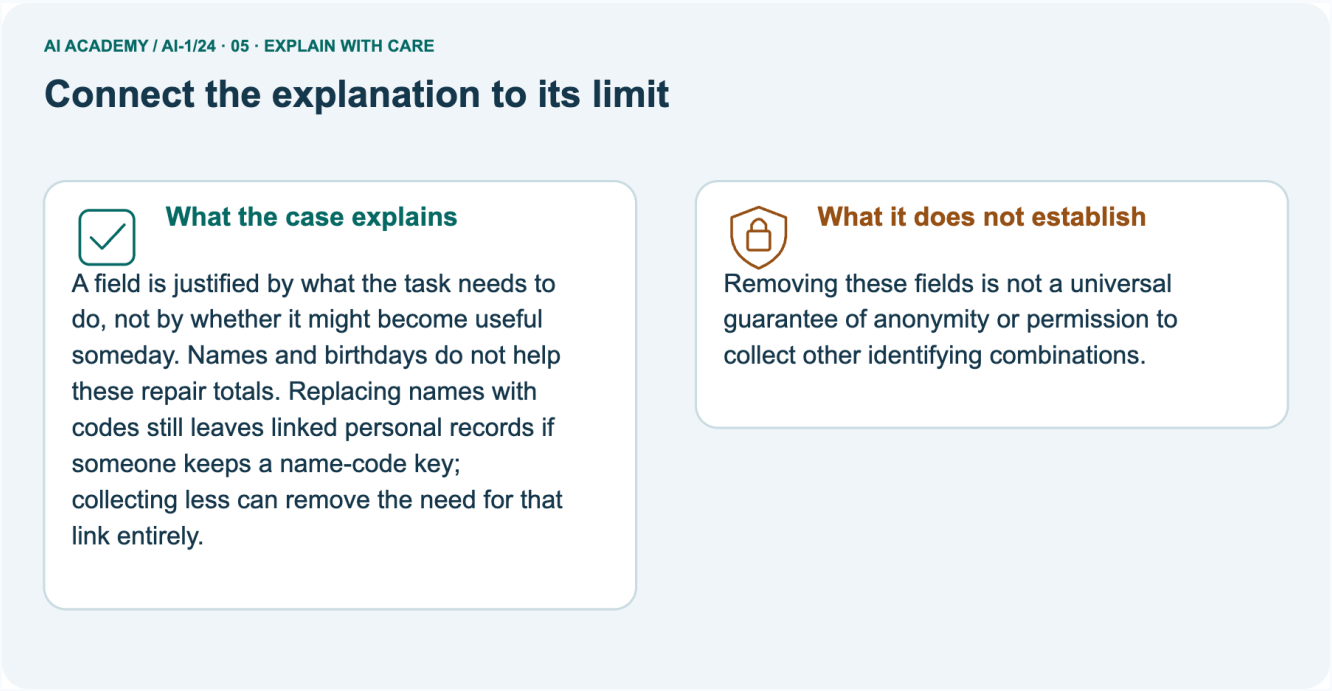
Expected result: A minimal contact **mechanism** may now be needed for that tool. Reason: The **purpose** changed; reassess necessity rather than restoring every old field.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **purpose** changed; reassess necessity rather than restoring every old field.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A field is justified by what the task needs to do, not by whether it might become useful someday. Names and birthdays do not help these repair totals.

Replacing names with codes still leaves linked personal records if someone keeps a name-code key; collecting less can remove the need for that link entirely.

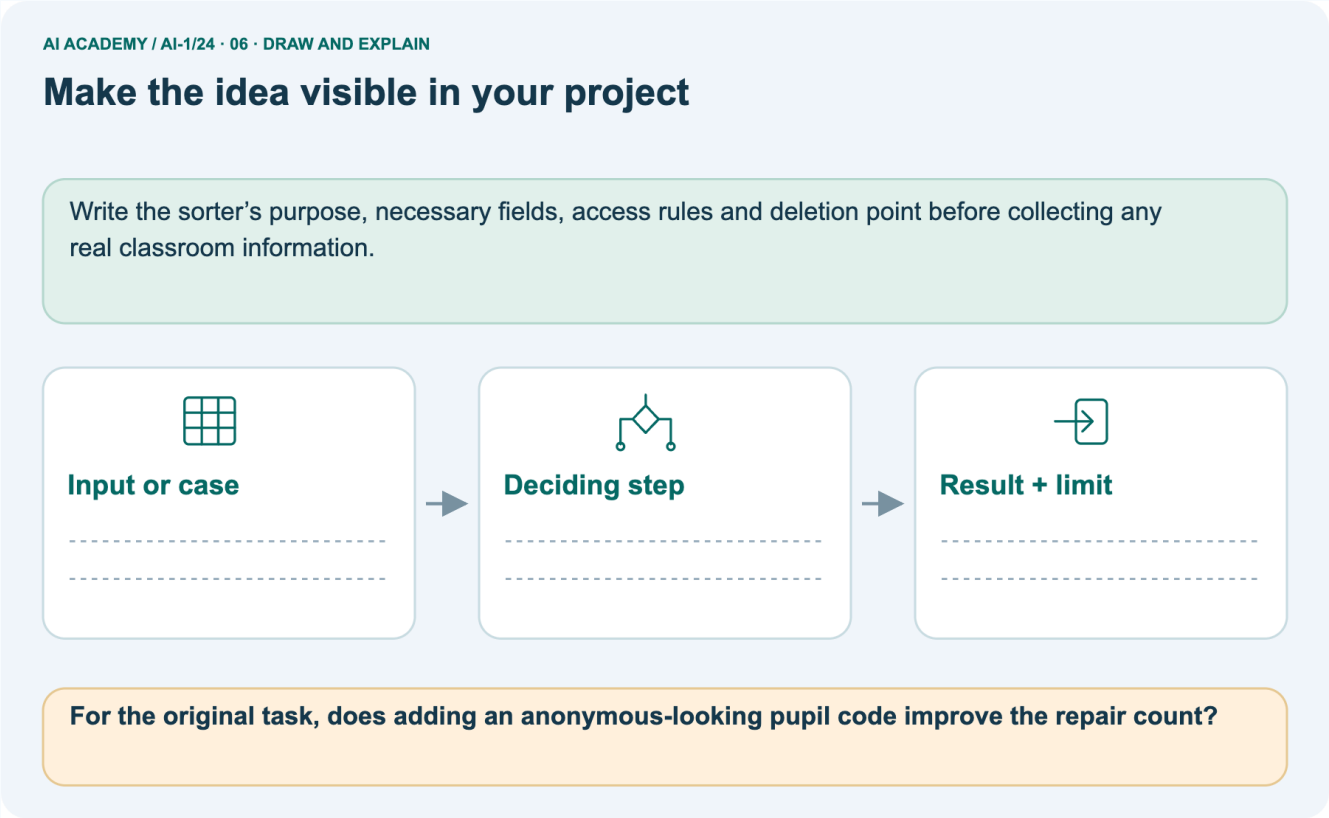
Limit: Removing these fields is not a universal guarantee of anonymity or permission to collect other identifying combinations.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Removing these fields is not a universal guarantee of anonymity or permission to collect other identifying combinations.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write the sorter's **purpose**, necessary fields, access rules and deletion point before collecting any real classroom information. Draw your **input**, deciding step, result and limit.

Supported question: For the original task, does adding an anonymous-looking pupil code improve the repair count?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. The totals need tool type and repair status, not pupil identity. A linkable code can also retain personal-data risk.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Privacy by Choosing Less: the case

02. Follow the deciding steps

03. A small information plan for a fictional vote

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.